A LOCALLY FINITE TYPE MORPHISM WHOSE FIBRE HAS DIMENSION ZERO AT A POINT IS LOCALLY QUASI-FINITE THERE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The existence and base change behaviour of the maximal open subspace on which a locally finite type morphism is locally quasi-finite. Invoked at the start of the counted proof (source lines 4266-4273) as cross-chapter pointers. Reduces the assertion to the membership of the given point in that open subspace.
- The statement that a morphism of algebraic spaces of relative dimension zero at a point is locally quasi-finite there. Invoked at the end of the counted proof (source lines 4274-4279) as a cross-chapter pointer. Gives local quasi-finiteness of the base-changed morphism at the point.

Source reference. Original reference spaces-morphisms-lemma-locally-quasi-finite-rel-dimension-0: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4279–4279. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-locally-quasi-finite-rel-dimension-0, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-item-loc-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4271–4271. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-item-loc-quasi-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference stacks-properties-remark-local-source-apply: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4270–4270. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-remark-local-source-apply, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Consider a cartesian diagram*

$$\begin{array}{ccc} U & \xleftarrow{p} & F \\ \downarrow & & \downarrow \\ \mathcal{X} & \longleftarrow & \mathrm{Spec}(k) \end{array}$$

where U is an algebraic space, k is a field, and $U \to \mathcal{X}$ is locally of finite type. Let $z \in |F|$ be such that $\dim_z(F) = 0$. Then, after replacing U by an open subspace containing $p(z)$, the morphism

$$U \longrightarrow \mathcal{X}$$

is locally quasi-finite.

Proof. Since $f : U \to \mathcal{X}$ is locally of finite type there exists a maximal open $W(f) \subset U$ such that the restriction $f|_{W(f)} : W(f) \to \mathcal{X}$ is locally quasi-finite, see Properties of Stacks, Remark stacks-properties-remark-local-source-apply (stacks-properties-item-loc-quasi-finite). Hence all we need to do is prove that $p(z)$ is a point of $W(f)$. Moreover, the remark referenced above also shows the formation of $W(f)$ commutes with arbitrary base change by a morphism which is representable by algebraic spaces. Hence it suffices to show that the morphism $F \to \mathrm{Spec}(k)$ is locally quasi-finite at z. This follows immediately from Morphisms of Spaces, Lemma spaces-morphisms-lemma-locally-quasi-finite-rel-dimension-0. □